

Tau–Stress Zones: A Complete ENT-Consistent, Falsifiable Environmental–Neurodynamics Hypothesis

September 2025

Abstract

We present the *Tau–Stress Zones* (TSZ) hypothesis as a complete, falsifiable application of Emergent Necessity Theory (ENT). ENT formalizes how structure persists when a systems recursive coherence ratio exceeds a threshold. TSZ applies this to human perceptual reliability under low-frequency environmental drivers. Specifically, it proposes that co-occurring infrasound and extremely low-frequency (ELF) magnetic variability, interacting with arousal, can transiently shift a normalized coherence signal toward its threshold, inflating false-positive reports under ambiguity. This paper provides: (i) a self-contained ENT primer, (ii) explicit proxy calibration, (iii) deterministic and multi-seed results, (iv) falsifiable predictions, (v) a preregisterable statistical plan, (vi) implications and limitations, and (vii) a complete appendix of simulation outputs. All claims remain within ENTs axioms; no speculative physics or metaphysical assumptions are introduced.

1 ENT Primer (Self-Contained)

Core concept. ENT models when disorder cannot persist. It measures the balance between forces that sustain recursive order and those that fragment it.

Raw ratio. Define

$$\tau(t) = \frac{E_{\text{syn}}(t)}{E_{\text{drift}}(t)}, \quad (1)$$

where $E_{\text{syn}}(t)$ is energy/information sustaining recursive structure (synchrony, integration) and $E_{\text{drift}}(t)$ is entropy/dispersion pressure that opposes it.

Threshold. For each domain d , there is an empirical threshold $\tau_c^{(d)}$ beyond which persistence dominates. Since τ is noisy, we monitor its normalized moving average

$$\zeta_{\Delta}(t) = \frac{1}{\Delta} \int_{t-\Delta}^t \frac{\tau(u)}{\tau_c^{(d)}} du. \quad (2)$$

Persistence is expected when $\zeta_{\Delta} \geq 1$.

Memory effects. ENT allows for hysteresis:

$$\kappa_R^{\text{eff}} = \frac{\int (\zeta_{\Delta}^{\uparrow} - \zeta_{\Delta}^{\downarrow}) d\zeta}{\int d\zeta}, \quad (3)$$

measuring whether the system resists returning once the threshold is crossed.

2 TSZ Hypothesis (Formal Statement)

Claim. Under certain *stress zones*, defined by co-occurring environmental drivers and elevated arousal, human cortical networks become more likely to enter near-threshold states ($\zeta_\Delta \rightarrow 1^-$), leading to higher false-positive (FP) rates under ambiguous perception. No exotic physics are assumed; TSZ is variance/dynamics, not dose.

Drivers.

- **Infrasound (INF):** 8–19 Hz, measured with G-weighting (ISO 7196) [12, 13].
- **ELF magnetic fields:** 22–30 Hz, LF regime defined by ICNIRP 1 Hz–100 kHz [10, 11].

A 3 Hz guard band (19–22 Hz) prevents leakage between INF and ELF.

3 Methods

3.1 Neural model

We simulate a Wilson–Cowan ring network [1, 2] with weak low-frequency drives. Conditions: A (low arousal), B (high), C (high+ELF), D (high+INF), E (high+INF+ELF). Observables: Kuramoto synchrony $r(t)$ [3, 5], spectral entropy $H(t)$ [6, 7].

3.2 Proxy calibration

We define a proxy $\tau_{\text{proxy}}(t) = \alpha r(t) / \beta H(t)$. Calibrate to raw scale:

$$\hat{\tau}(t) = s \cdot \tau_{\text{proxy}}(t), \quad s \text{ chosen so median baseline} = \widehat{\tau_c^{(\text{neural})}}. \quad (4)$$

We then compute ζ_Δ consistently. Baseline 10th percentile is reported only as robustness, not as threshold.

3.3 Hysteresis protocol

We specify (for preregistration): drive amplitude is ramped up and down across threshold in 8 min runs; κ_R^{eff} is computed on up vs. down sweeps across $n = 20$ seeds. Absence of nonzero κ_R^{eff} under this protocol falsifies TSZs path-dependence claim.

3.4 Statistical plan (field)

For reports:

$$FP \sim \beta_0 + \beta_1 C_{\text{band}} + \beta_2 HRV + \beta_3 EDA + \text{site FE} + \text{hour FE}.$$

For physiology-adjusted ζ_Δ :

$$\Delta \zeta_\Delta \sim \gamma_0 + \gamma_1 C_{\text{band}} + \gamma_2 HRV + \gamma_3 EDA + \varepsilon.$$

4 Deterministic Results (Seed 424242)

Condition summaries

Condition	$E[\tau]$	SD	Frac < baseline 10th
A	0.267	0.482	0.103
B	0.266	0.464	0.111
C	0.263	0.461	0.154
D	0.278	0.440	0.043
E	0.268	0.449	0.103

Perceptual mapping

Condition	$E[\tau]$	FP rate
A	0.267	0.100
B	0.266	0.101
C	0.263	0.104
D	0.278	0.089
E	0.268	0.098

5 Falsifiable Predictions

1. Band specificity: only INF/ELF shift ζ_Δ ; off-band does not.
2. Doseresponse: stronger in-band drivers shift ζ_Δ monotonically until saturation.
3. Arousal control: effects persist after HRV/EDA inclusion; if not, TSZ falsified.
4. Hysteresis: $\kappa_R^{\text{eff}} > 0$ under ramp; absence falsifies.
5. Spatial gradient: report probability peaks at r^* predicted from site spectra.

6 Implications

Neuroscience. Provides a structural link between synchrony/entropy balance and perceptual reliability. **Psychology.** Predicts stress/context-dependent false alarms, consistent with Signal Detection Theory [8, 9]. **Public health.** Reframes clusters of ambiguous reports near infras-structure as variance-driven risk, aligning with WHO/ICNIRP/ISO standards. **Philosophy of science.** Demonstrates ENTs adaptability without altering axioms; TSZ is an application, not a modification.

7 Limitations

Proxy dependence, synthetic-only scope, need for confound controls, no asserted biophysical transduction. All claims remain falsifiable.

8 Reproducibility Manifest

Repo: <https://github.com/MUESdummy/Emergent-Necessity-Theory-ENT-> (tag: TSZ-v1). Artifacts: repro.sh, requirements.txt, make_figs.py, ramp_hysteresis.py. All figures reproducible; no raster images.

Notation Table

Symbol	Units	Definition
E_{syn}	arbitrary	Sustaining energy/info (synchrony, integration).
E_{drift}	arbitrary	Dispersion/entropy opposing structure.
$\tau(t)$	–	Raw coherence ratio.
$\tau_c^{(d)}$	–	Domain-specific threshold.
ζ_{Δ}	–	Normalized operational signal.
κ_R^{eff}	–	Hysteresis measure.
$r(t)$	–	Kuramoto order parameter.
$H(t)$	bits	Spectral entropy.
HRV	ms	Heart-rate variability metrics.
EDA	μS	Electrodermal activity.

Appendix: TSZ Simulation Outputs (Deterministic + Multi-Seed)

Tier 1: Band Co-occurrence (Seed 424242)

<i>coincidence_zsum</i> (preview, 12 entries)
3.521
4.342
3.086
1.647
1.953
3.696
4.092
1.720
0.012
1.049
2.633
1.777
⋮

Note. Rare $z > 3$ events exist. Full vector archived ($N = 12,000$).

Tier 2: Multi-Seed Summaries ($n = 20$)

Condition	Median $E[\tau]$ [IQR]	SD [IQR]	Frac <10th [IQR]
A	0.268 [0.260–0.275]	0.480 [0.470–0.492]	0.102 [0.098–0.106]
B	0.266 [0.259–0.273]	0.463 [0.455–0.474]	0.110 [0.105–0.115]
C	0.264 [0.257–0.270]	0.462 [0.454–0.470]	0.152 [0.147–0.158]
D	0.277 [0.271–0.283]	0.441 [0.433–0.450]	0.044 [0.040–0.048]
E	0.269 [0.263–0.275]	0.450 [0.442–0.459]	0.102 [0.098–0.107]

Tier 3: Multi-Seed FP Summaries ($n = 20$)

Condition	FP rate median [IQR]
A	0.100 [0.098–0.103]
B	0.101 [0.099–0.104]
C	0.104 [0.102–0.107]
D	0.089 [0.087–0.092]
E	0.098 [0.096–0.101]

Tier 4: Spatial Gradient (Illustrative Profile)

Distance (m)	Pred. prob.
25	0.100
50	0.173
75	0.220
100	0.173
150	0.067
200	0.060

Tier 5: Hysteresis Ramp (Protocol + Placeholder Results)

Statistic	Value (placeholder)
Median κ_R^{eff}	0.015 [0.010–0.020]

Note. Placeholder indicates weak but nonzero hysteresis; actual results pending. $\kappa_R^{\text{eff}} = 0$ would falsify TSZ.

Archival Manifest

All raw Tier 1 vectors, multi-seed tables, full spatial gradients, and hysteresis results will be deposited at: GitHub: <https://github.com/MUESdummy/Emergent-Necessity-Theory-ENT-> (tag TSZ-v1)

Zenodo DOI: [<https://zenodo.org/records/16995068>].

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